Attachment C · HMI screen mock-ups · Rev 1

Solarization of Villa Somalia · SCADA Design Report Rev 1 · Blueflag Energy · drawing set BFE/PR/26/01

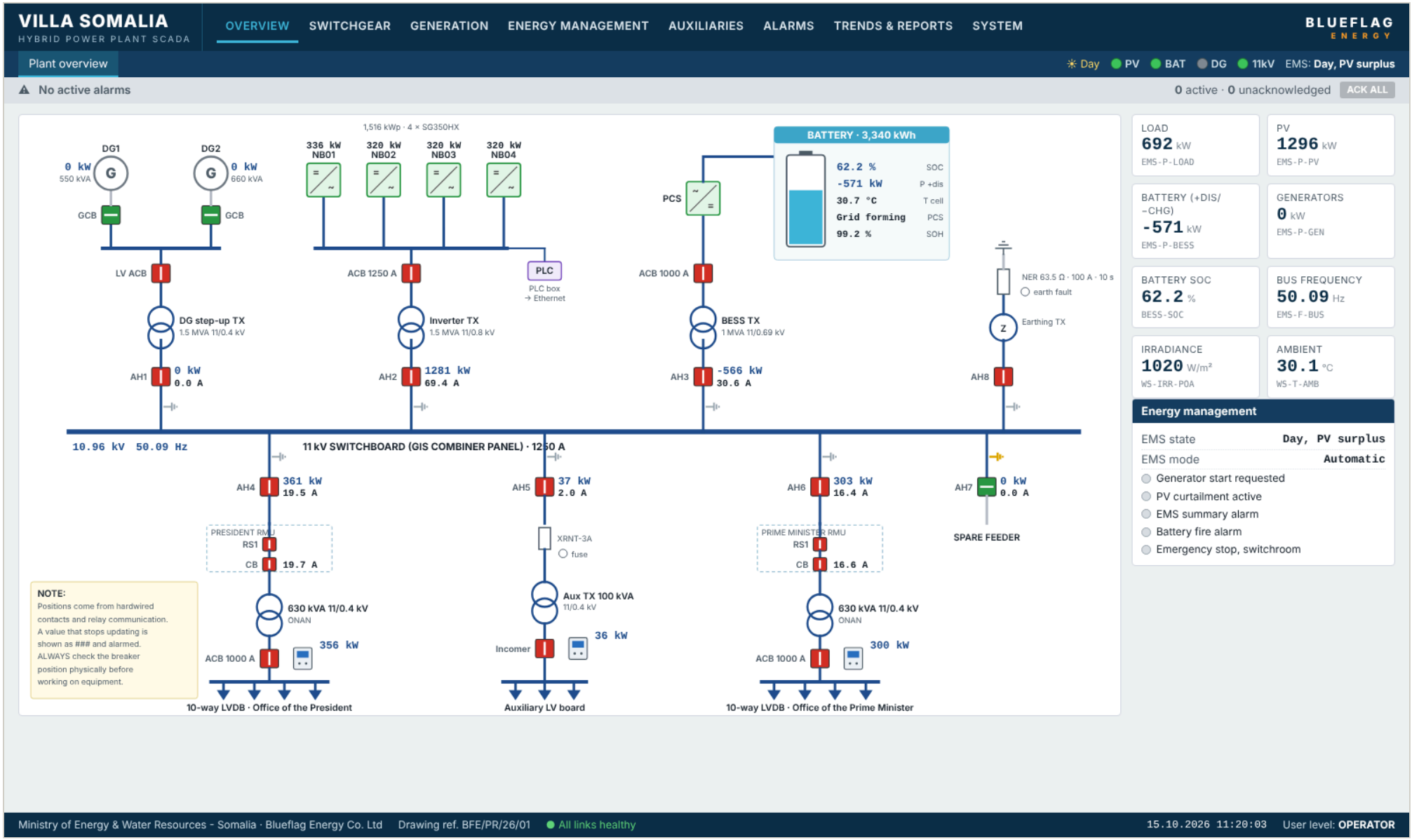
> These screens are mock-ups from the HMI demonstration, running on a simulator of the plant mapped onto every signal of the device and I/O list (Attachment B). They are submitted for approval of layout and content. The final screens are built in SICAM SCC to the same layout and content before the factory test.

Conventions: breaker closed red and open green, as on the screens already used on site; live conductors dark blue, dead grey; alarm levels 1 red, 2 orange and 3 yellow; values from a device that has stopped communicating shown as ### in magenta, never as live. Every command opens a confirmation step, and controls a user may not operate are disabled for that role.

C.1	Plant overview
C.2	11 kV switchboard
C.3	Office transformer station (President)
C.4	PV system
C.5	Battery
C.6	Generators
C.7	Energy management
C.8	Auxiliaries
C.9	Communication status
C.10	I/O list
C.11	Plant overview during a battery fire
C.12	Alarm list during a battery fire

C.1 Plant overview

The opening screen: a live single-line diagram of the whole plant. Generators, PV, battery, the 11 kV switchboard AH1 to AH8, both office stations and the auxiliary supply, with power at each point, breaker positions, the EMS state and summary values with their tags.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.2 11 kV switchboard

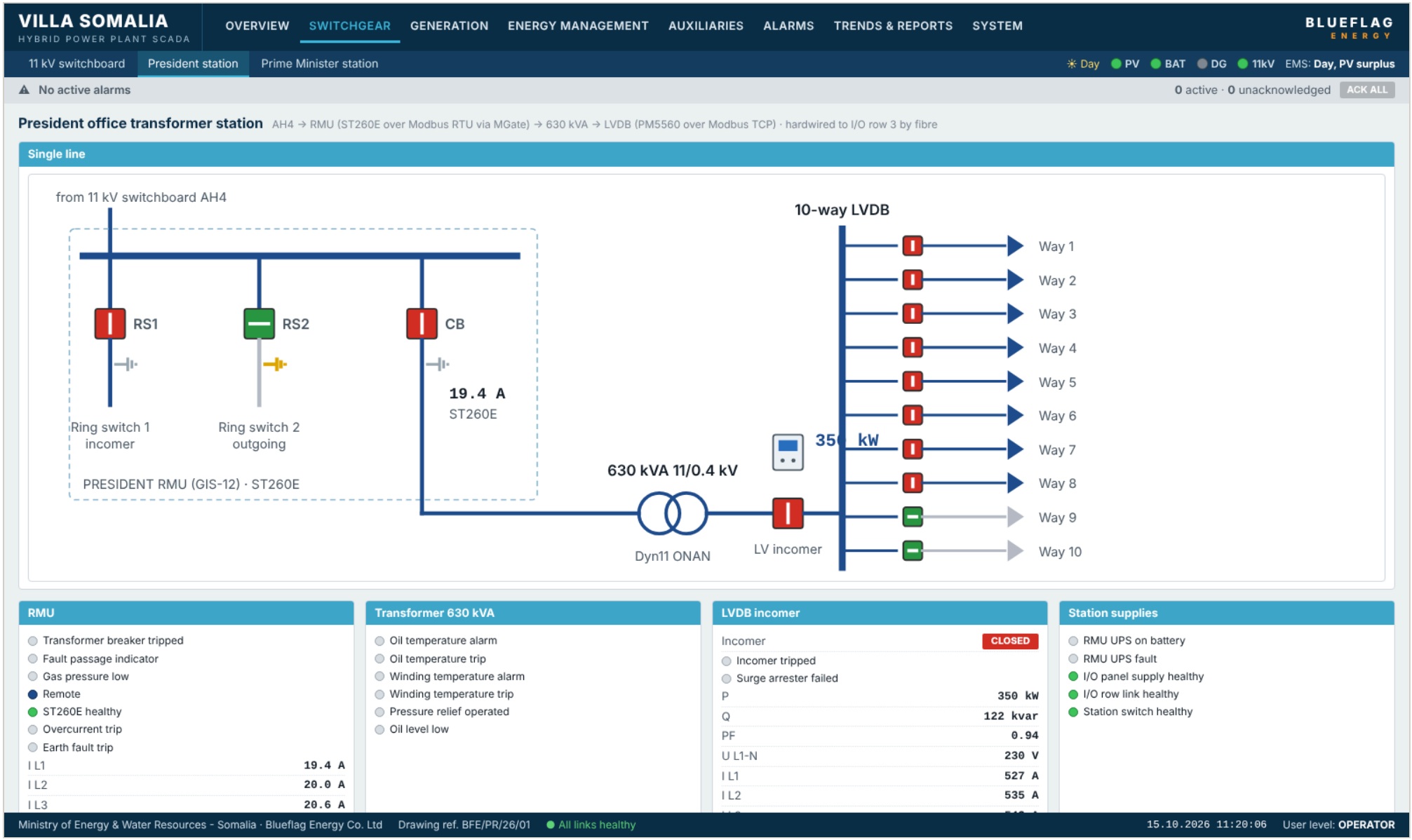
One column per panel AH1 to AH8: breaker and earth switch positions, local/remote, gas pressure, relay health and trips, measurements and energy. An Operator opens and closes breakers by select and confirm; interlocks are shown in the breaker faceplate.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.3 Office transformer station (President)

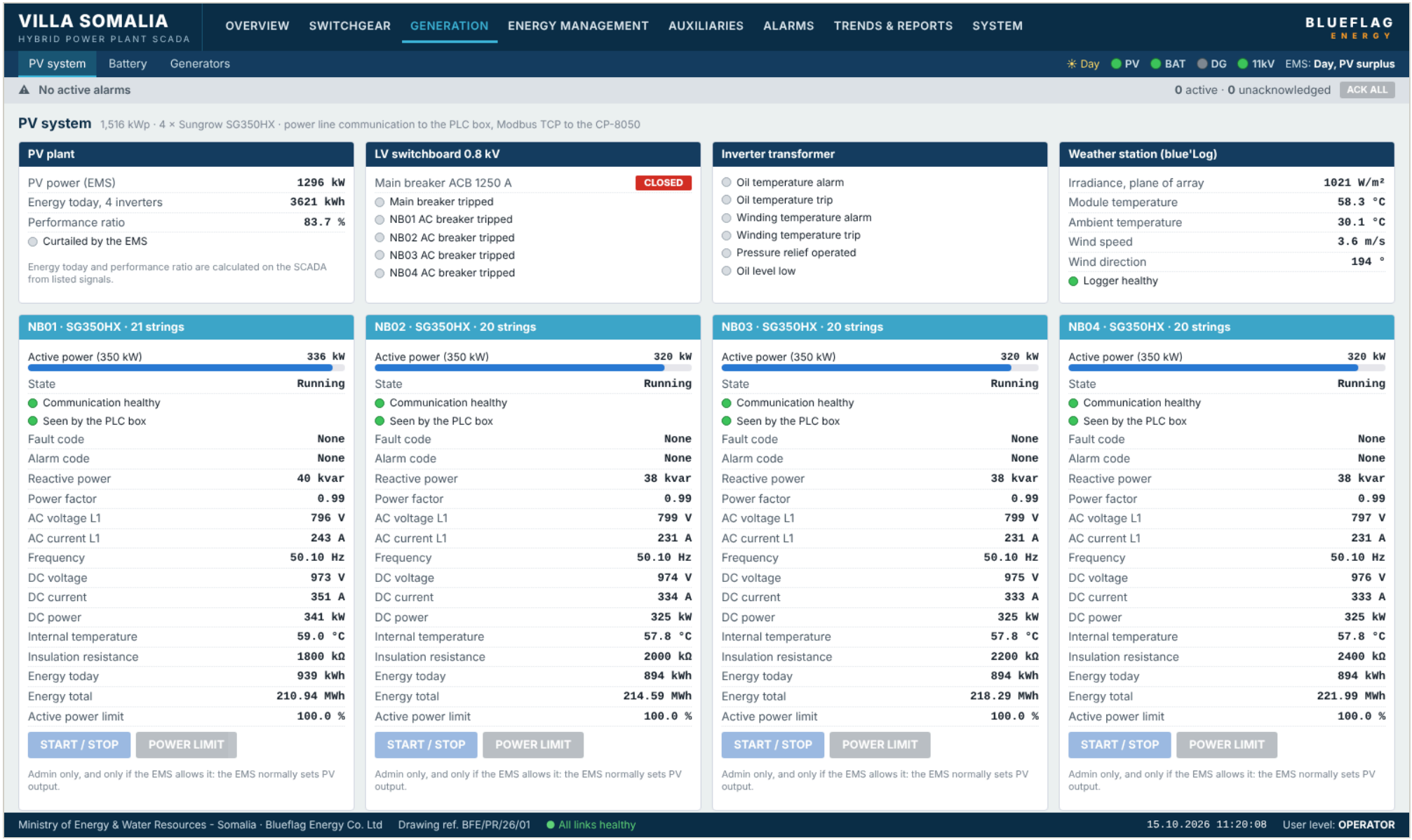
RMU switches and transformer breaker with the ST260E relay, 630 kVA transformer alarms, the 10-way LVDB with its incomer and meter values, and the station supplies. The Prime Minister station screen is the same.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.4 PV system

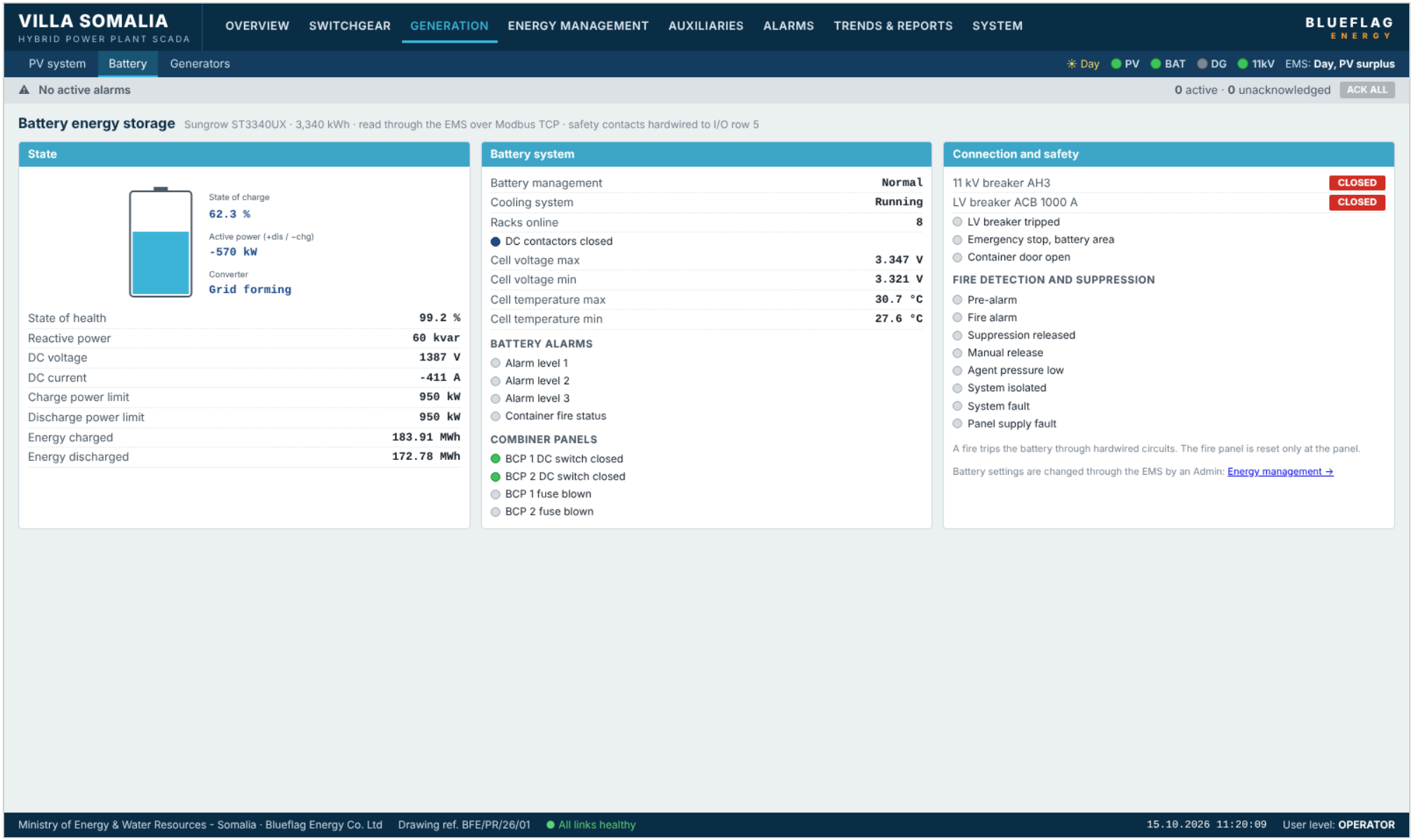
The four SG350HX inverters as read through the PLC box: power, voltages, currents, temperature, insulation, state and fault codes; the PV LV switchboard and AC breakers, the inverter transformer, and the weather station.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.5 Battery

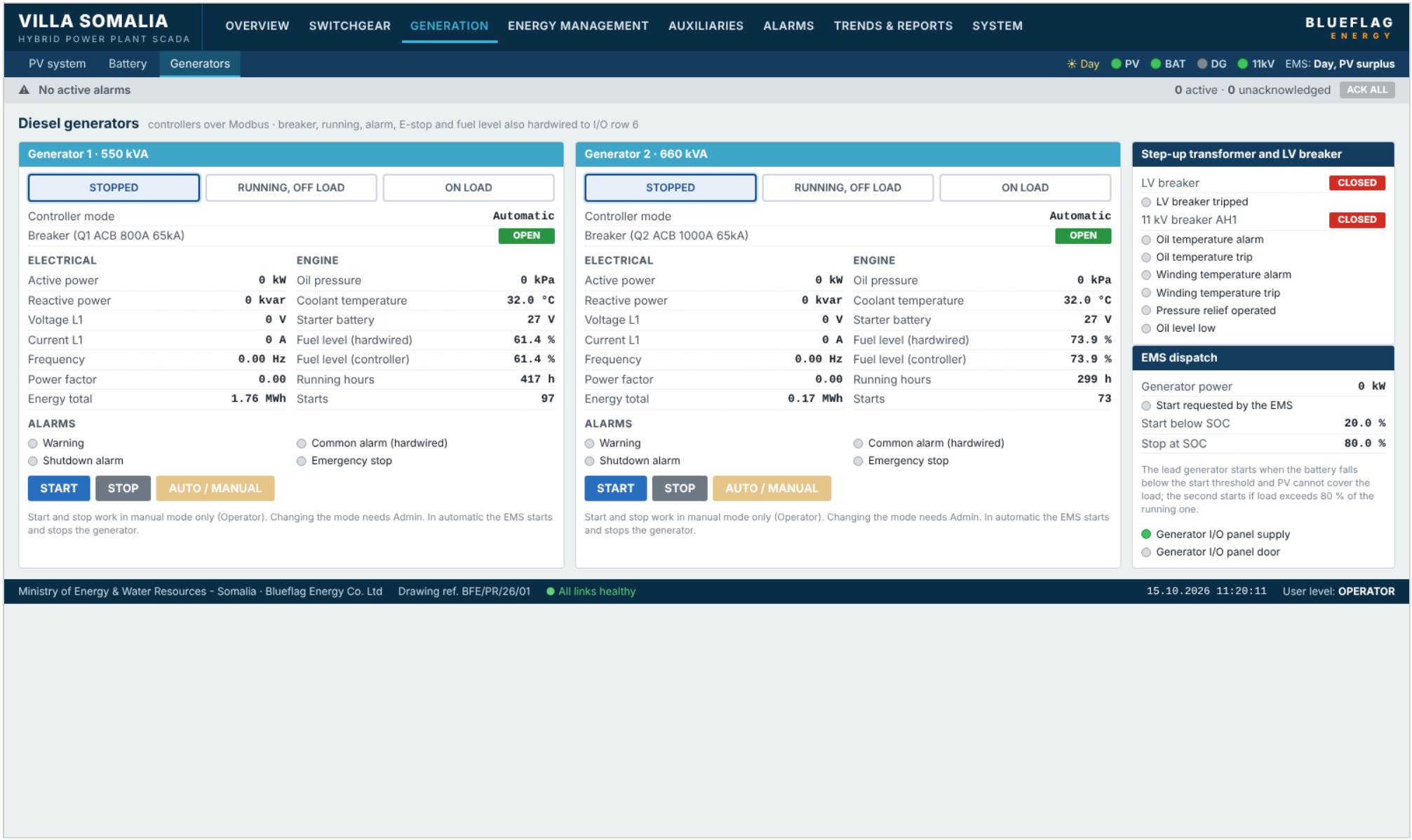
The ST3340UX battery as read through the EMS: state of charge and health, power, charge and discharge limits, cell values, alarms, combiner panels, and the hardwired fire panel, emergency stop and breaker signals.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.6 Generators

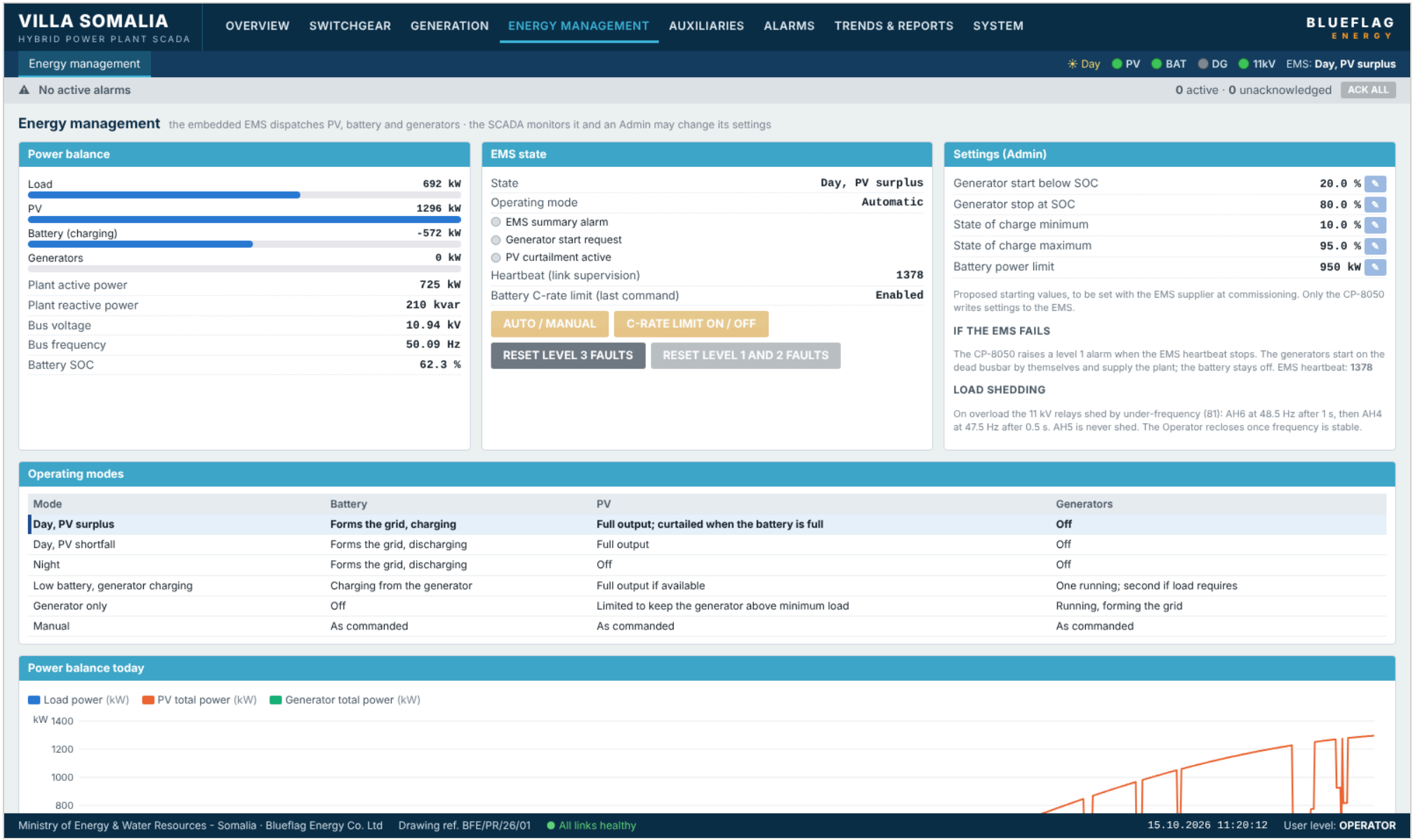
DG1 550 kVA and DG2 660 kVA: state, electrical and engine values, fuel level hardwired and from the controller, running hours and starts. Start and stop work in manual mode only; changing the mode needs the Admin role.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.7 Energy management

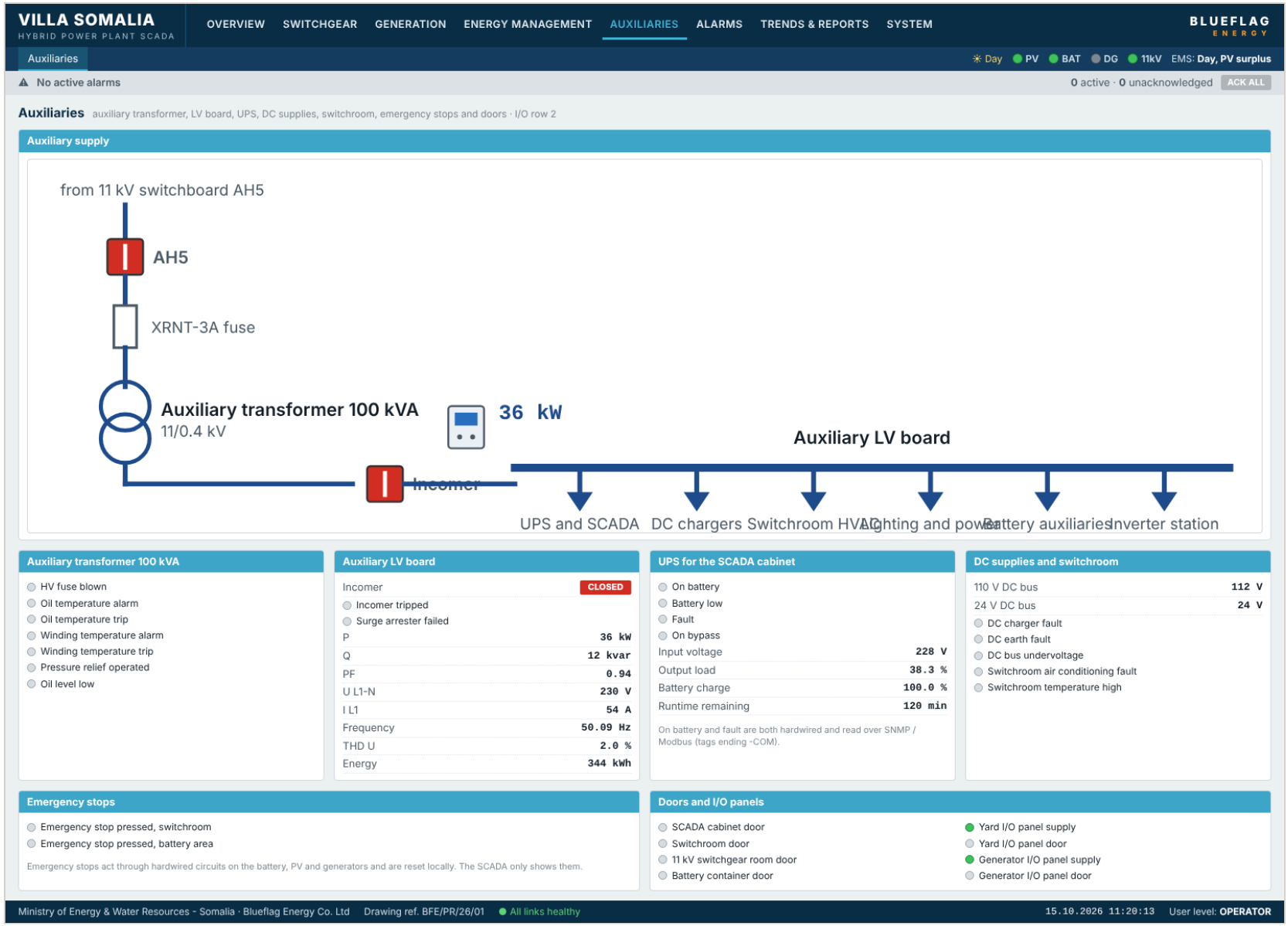
EMS state and mode, the power balance, the operating modes with the current one highlighted, and the Admin settings: generator start below 20 % and stop at 80 % state of charge. Also the fallback if the EMS fails and the load-shedding order.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.8 Auxiliaries

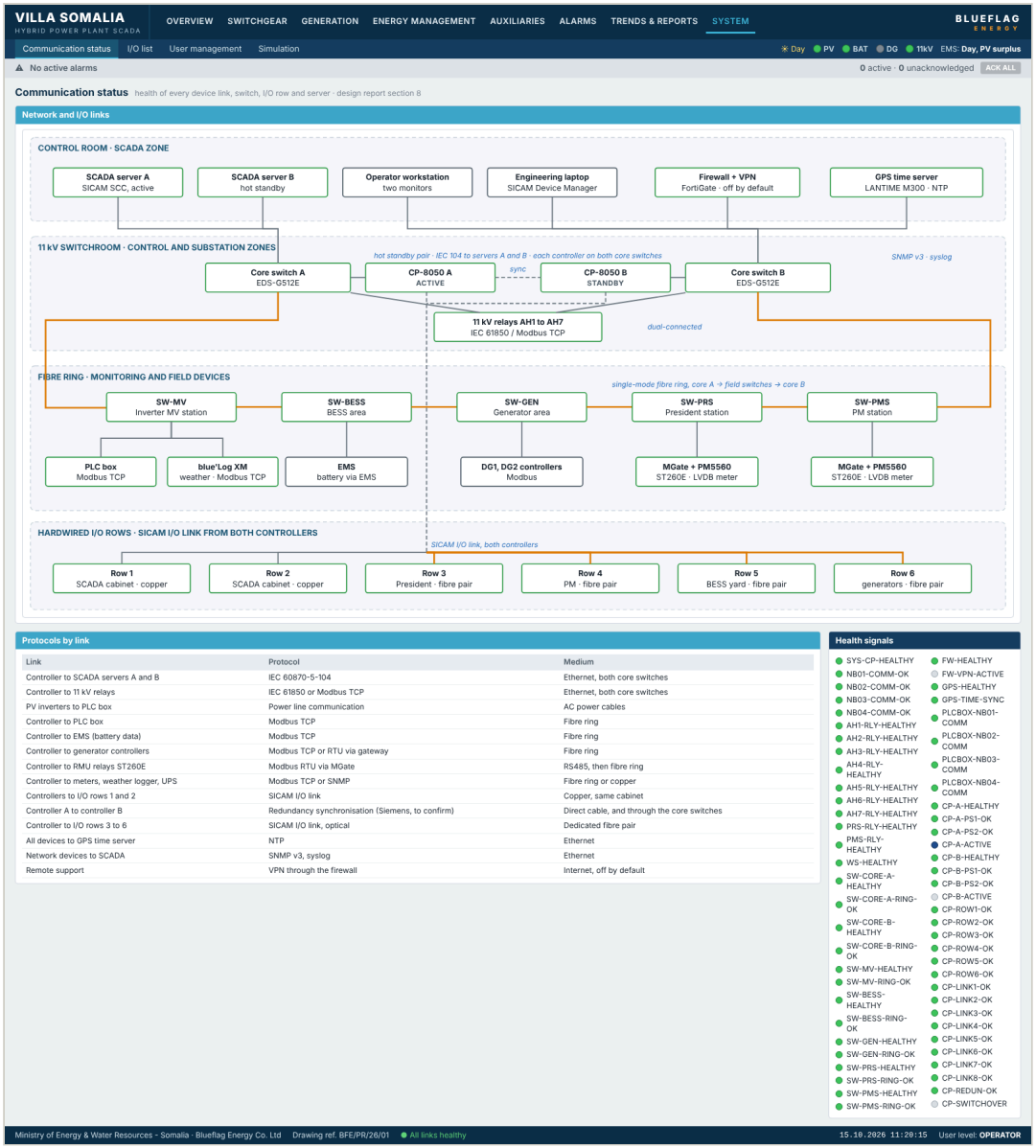
Auxiliary transformer and LV board, the SCADA UPS, DC supplies, switchroom conditions, emergency stops and doors. Emergency stops are hardwired; the SCADA only shows them.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.9 Communication status

Live health of every link: the hot standby controller pair with its active and standby roles, both core switches, the fibre ring and field switches, field devices and the six I/O rows, with the protocol of each link.

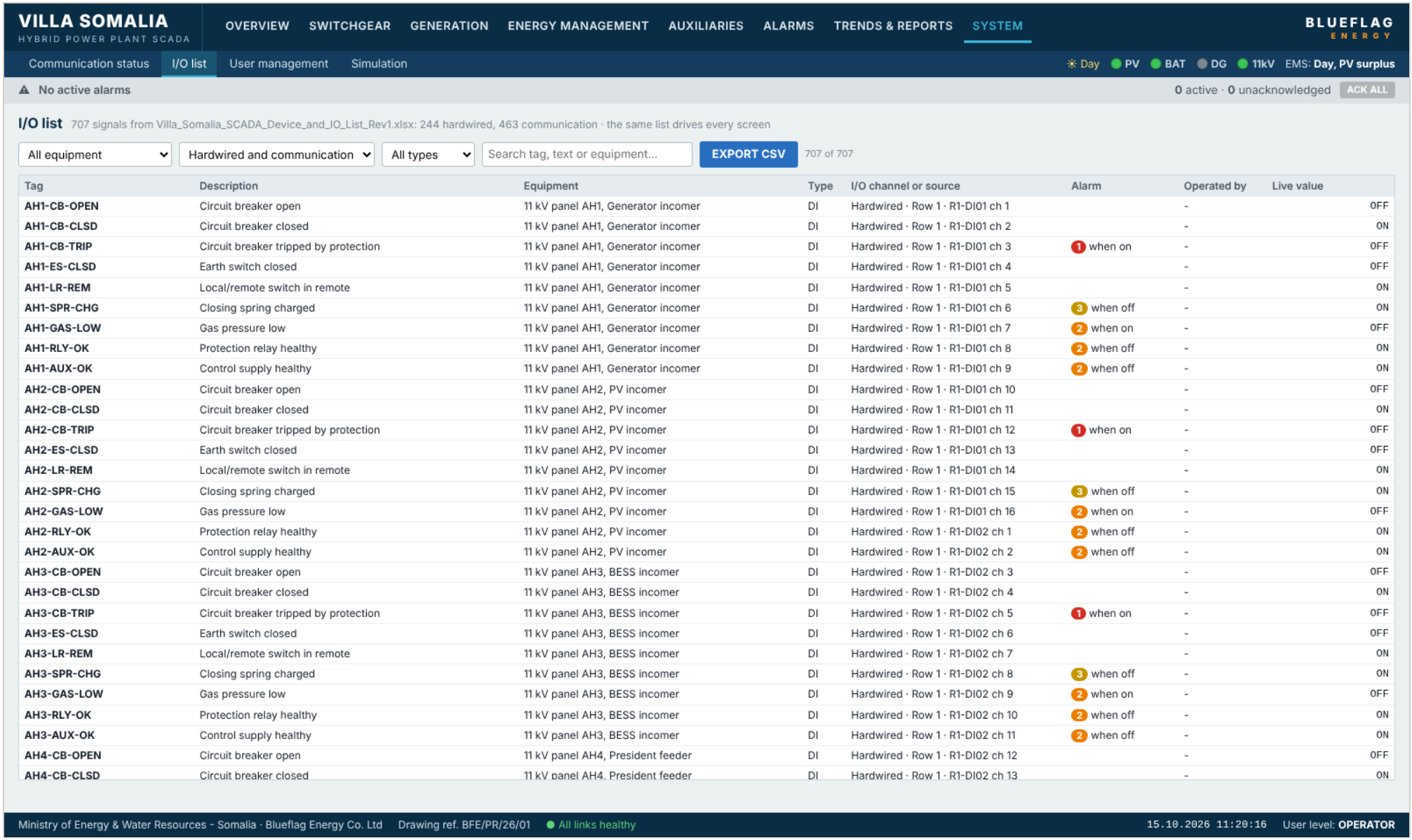


Link	Protocol	Medium
Controller to SCADA servers A and B	IEC 60870-5-104	Ethernet, both core switches
Controller to 11 kV relays	IEC 61850 or Modbus TCP	Ethernet, both core switches
PV inverters to PLC box	Power line communication	AC power cables
Controller to PLC box	Modbus TCP	Fibre ring
Controller to EMS (battery data)	Modbus TCP	Fibre ring
Controller to generator controllers	Modbus TCP or RTU via gateway	Fibre ring
Controller to RMU relays ST260E	Modbus RTU via MGate	RS485, then fibre ring
Controller to meters, weather logger, UPS	Modbus TCP or SNMP	Fibre ring or copper
Controllers to I/O rows 1 and 2	SICAM I/O link	Copper, same cabinet
Controller A to controller B	Redundancy synchronisation (Siemens, to confirm)	Direct cable, and through the core switches
Controller to I/O rows 3 to 6	SICAM I/O link, optical	Dedicated fibre pair
All devices to GPS time server	NTP	Ethernet
Network devices to SCADA	SNMP v3, syslog	Ethernet
Remote support	VPN through the firewall	Internet, off by default

Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.10 I/O list

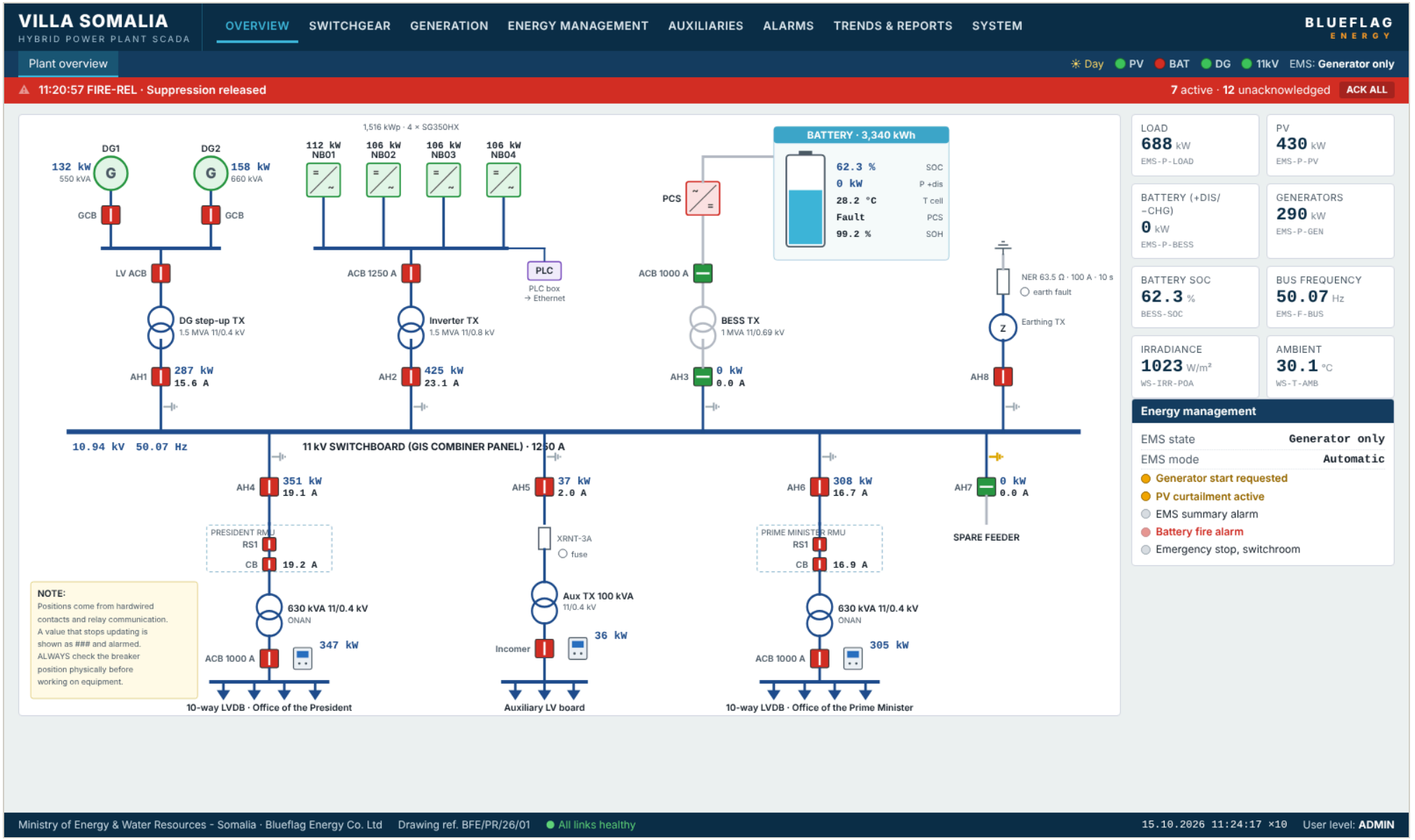
Every signal of the device and I/O list (Attachment B) with its tag, I/O row, module and channel or protocol source, alarm level and live value, with filters and export.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.11 Plant overview during a battery fire

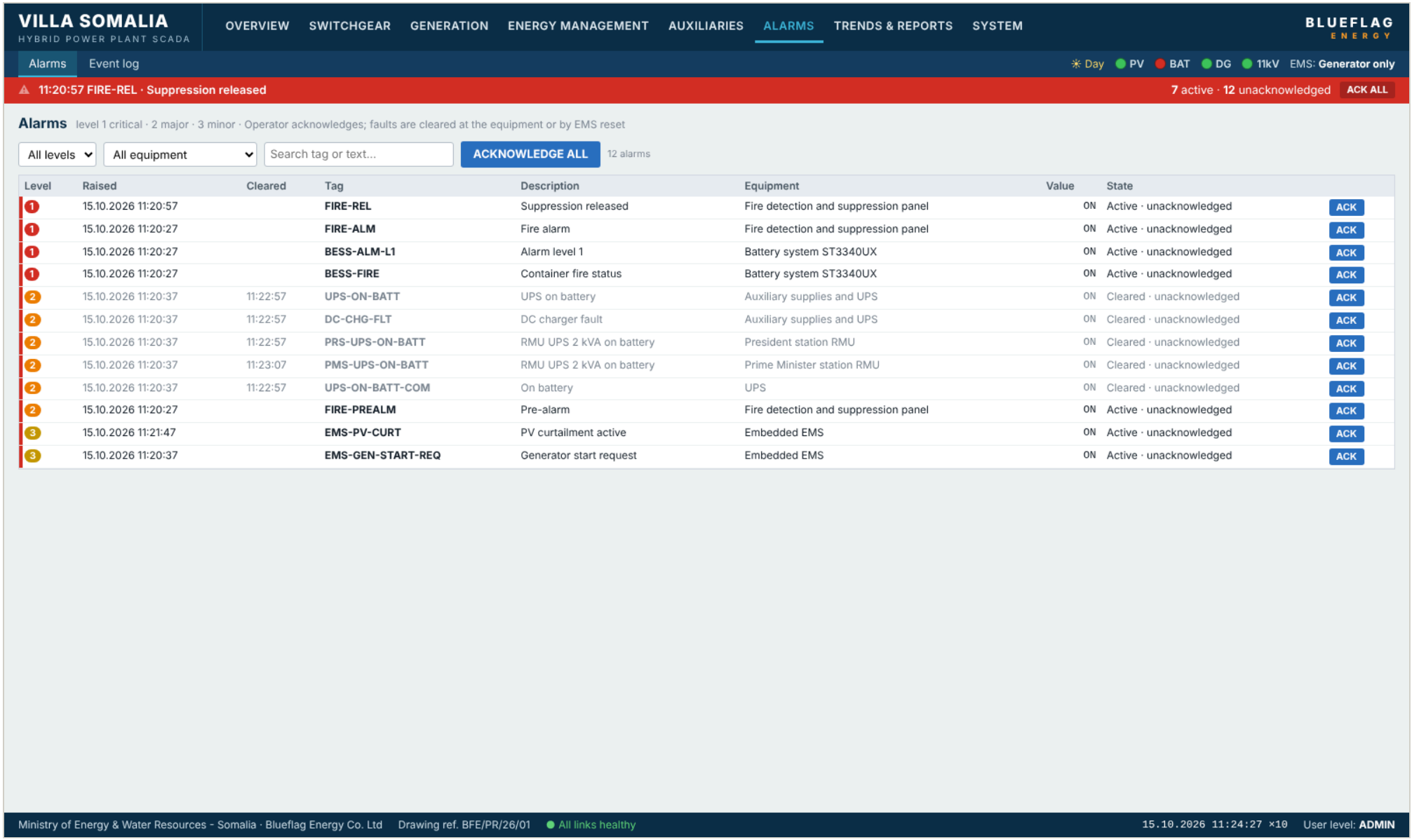
The fire panel has tripped the battery by its hardwired circuit. The generators have started, the busbar has been restored with AH8 closed first, and both offices are supplied again; PV runs limited to keep the generators above minimum load.



Mock-up from the HMI demonstration · Attachment C to the SCADA Design Report Rev 1

C.12 Alarm list during a battery fire

The alarms raised by the fire, with level, time, tag, equipment and state. Each alarm is acknowledged singly or all at once; fire alarms clear only when reset at the fire panel.



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